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LMatchNet-V: Physics-Seeded Learning for Vendor-Aware RF L-Section Matching with Adaptive Safe-Library Search

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***Abstract*—Discrete vendor-component selection remains a bottleneck in classical L-section RF impedance matching because analytical synthesis yields continuous values, whereas practical components are quantized, lossy, and frequency-dependent. This paper presents LMatchNet-V, a physics-seeded learning framework for vendor-aware single-frequency matching. An analytical seed is refined by a residual model trained using load, frequency, pad-capacitance, branch, and synthetic finite-loss information, while inference uses only pre-selection load and design quantities. Final component selection is performed through vendor-table circuit-model recomputation rather than direct ML prediction. In a verification set containing 20,000 branch-specific cases, 5,514 were feasible under exhaustive safe-library search. The fixed no-Q 5×5 search recovered 88.47% of the feasible frontier using 25 vendor-pair evaluations. Under feasible-frontier adaptive accounting, 7×7 expansion followed by exhaustive fallback recovered all 5,514 feasible cases, with a median of 25 unique evaluations. These results demonstrate effective discrete vendor-component search localization while retaining vendor-model verification as the final decision authority.**

***Keywords*—RF impedance matching, vendor-aware component selection, discrete passive components, physics-informed machine learning, residual learning, design automation.**

I. Introduction

Impedance matching between a radio-frequency (RF) source and its load is a basic requirement in narrowband transceiver design. Poor matching reflects signal power, reduces gain, and can destabilize power amplifiers. For single-frequency applications, the two-element L-section network remains attractive for its minimal topology and closed-form synthesis [1].

Classical synthesis yields continuous L and C values for a given load and reference impedance. In practice, the analytical pair must be mapped to discrete vendor components that are quantized, frequency-dependent, and lossy. Nearest catalog values may yield poor S_{11} once vendor models are applied. Exhaustive search can become costly under repeated or larger-library evaluation. The challenge is preserving acceptable S_{11} after projection onto a discrete, frequency-dependent library.

Machine learning has been applied to impedance matching through deep-learning-based prediction and adaptation [2], [3], and its broader use in antenna design and optimization has been reviewed in [4]. Related RF and analog design studies have investigated Bayesian optimization [5], regression-based microwave surrogate modeling [6], AI-assisted microwave optimization [7], large-language-model-assisted analog design [8], and recurrent-network surrogate modeling [9]. Adaptive L-network control [10], shallow-learning impedance matching [11], and deep-learning-assisted active-RF layout optimization [12] have also been reported. However, these approaches do not directly guarantee feasible selection from a finite, frequency-dependent vendor catalog.

This paper presents **LMatchNet-V**, a physics-seeded learning framework for vendor-aware L-section impedance matching. As shown in **Fig. 1**, classical synthesis provides a seed; a residual model predicts corrections; and the corrected point anchors a compact vendor search. Candidate pairs are evaluated using interpolated vendor impedance tables, and the best evaluated local candidate is selected. Machine learning acts as a search-space filter, not a replacement for circuit evaluation.

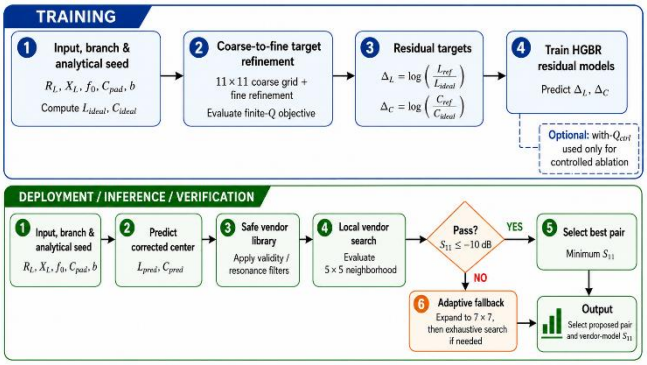


Fig. 1. Simplified overview of the proposed LMatchNet-V methodology for vendor-aware RF L-section matching.

The model uses only pre-selection quantities: load resistance, load reactance, frequency, pad capacitance, and branch. No synthetic loss-control features are needed at inference. An adaptive fallback expands to exhaustive search when needed.

The main contributions of this work are as follows:

i. Physics-seeded residual learning that refines analytical L-section synthesis.
ii. Vendor-aware search protocol: learning proposes the center; vendor-model S_{11} selects the final pair.
iii. A deployable no-Q formulation that avoids synthetic loss-control at inference.

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iv. Verification on generated load states with Murata impedance tables, against analytical, random, elite-guided, and exhaustive baselines.

The study uses vendor-table circuit-model evaluation, not hardware measurement.

II. Methodology

A. Problem Formulation and Vendor-Model Evaluation

This work considers single-stage L-section impedance matching for low-resistance loads satisfying $R_L < Z_0$, where $Z_0 = 50\ \Omega$ and

$$Z_{\text{load}} = R_L + jX_L. \tag{1}$$

For each case, the known quantities are R_L, X_L, operating frequency f_0, input pad capacitance C_{pad}, and branch topology. For the considered low-resistance regime, the series element is placed adjacent to the load and the shunt element at the input node. In the L_{LC} branch, the series element is an inductor and the shunt element is a capacitor; in the L_{CL} branch, these roles are reversed.

For a vendor candidate pair (L_i, C_j), the frequency-dependent complex impedances are linearly interpolated at f_0, giving $Z_{\text{ind},i}(f_0)$and $Z_{\text{cap},j}(f_0)$. The branch-specific series and shunt impedances are

$$L_{LC}\colon\ Z_s = Z_{\text{ind},i}(f_0),\ Z_p = Z_{\text{cap},j}(f_0), \tag{2}$$

$$L_{CL}\colon\ Z_s = Z_{\text{cap},j}(f_0),\ Z_p = Z_{\text{ind},i}(f_0). \tag{3}$$

The network impedance before including the input pad capacitance is

$$Z_{\text{net}} = \left[\frac{1}{Z_{\text{load}} + Z_s} + \frac{1}{Z_p}\right]^{-1}. \tag{4}$$

The input pad capacitance is included as a shunt admittance:

$$Y_{\text{in}} = \frac{1}{Z_{\text{net}}} + j2\pi f_0 C_{\text{pad}},\ Z_{\text{in}} = \frac{1}{Y_{\text{in}}}. \tag{5}$$

The reflection coefficient and S_{11} magnitude in decibels is

$$\Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0},\ S_{11,\text{dB}} = 20\log_{10}|\Gamma|. \tag{6}$$

A candidate passes when $S_{11,\text{dB}} \le -10$ dB. Exhaustive evaluation over the filtered safe library defines the vendor-feasibility benchmark.

B. Physics-Seeded Residual Learning

Classical low-resistance L-section synthesis provides continuous seed values L_{ideal} and C_{ideal}. Let $r = R_L/Z_0$, $x_L = X_L/Z_0$, and $\omega_0 = 2\pi f_0$. The seed is obtained from

$$x_t = \pm\sqrt{r - r^2},\ X_s = (x_t - x_L)Z_0,\ B_p = \frac{x_t}{rZ_0},$$

$$x_t > 0\colon\ (L_{\text{ideal}}, C_{\text{ideal}}) = \left(\frac{X_s}{\omega_0}, \frac{B_p}{\omega_0}\right), \tag{7}$$

$$x_t < 0\colon\ (L_{\text{ideal}}, C_{\text{ideal}}) = \left(-\frac{1}{\omega_0 B_p}, -\frac{1}{\omega_0 X_s}\right).$$

The positive and negative solutions correspond to the $L_{LC}$and L_{CL} branches, respectively. These values serve only as physics-based search coordinates; final evaluation follows the vendor topology in (2)–(6).

The analytical seed does not include vendor discreteness, frequency-dependent loss, pad capacitance, or self-resonance. Training targets are therefore generated using a two-stage multiplicative refinement. The coarse stage evaluates all combinations of

$$s_L, s_C \in \{0.40, 0.50, 0.63, 0.78, 0.90, 1.00, 1.12, 1.28, 1.55, 1.90, 2.40\}.$$

$$L = s_L L_{\text{ideal}},\ C = s_C C_{\text{ideal}}. \tag{8}$$

The best coarse region is refined using a 21-point local search within a $\pm 15\%$span. Candidates outside $0.05\ \text{nH} \le L \le 100\ \text{nH}$or $0.005\ \text{pF} \le C \le 50\ \text{pF}$are discarded.

During target generation, the inductor is represented by a finite-Q series model, while the capacitor is treated as ideal:

$$R_{s,L} = \frac{\omega_0 L}{Q_{\text{ctrl}}},\ Z_{\text{ind}}^{(Q)} = R_{s,L} + j\omega_0 L,\ Z_{\text{cap}}^{(Q)} = \frac{1}{j\omega_0 C}. \tag{9}$$

The pad capacitance is included through (5). The synthetic variable Q_{ctrl} is sampled over $30 \le Q_{\text{ctrl}} \le 100$ and is used only for target generation, not as a deployable model input.

The local-refinement objective is

$$J = \max\left(0, -S_{11,\text{dB}}\right) - 0.20\sqrt{(\ln s_L)^2 + (\ln s_C)^2}. \tag{10}$$

The first term rewards lower reflection, while the second discourages unnecessary displacement from the analytical seed. The candidate maximizing $J$is stored as $(L_{\text{ref}}, C_{\text{ref}})$.

Final verification is performed using the frequency-dependent vendor impedance tables.

The supervised targets are natural-log residuals:

$$\Delta_L = \ln\left(\frac{L_{\text{ref}}}{L_{\text{ideal}}}\right),\ \Delta_C = \ln\left(\frac{C_{\text{ref}}}{C_{\text{ideal}}}\right). \tag{11}$$

Two histogram-gradient-boosting regressors predict $\widehat{\Delta}_L$and $\widehat{\Delta}_C$, respectively. The deployable no-Q feature vector is

$$x_{\text{noQ}} = \left[\frac{R_L}{Z_0},\ \frac{X_L}{Z_0},\ \log_{10}\left(\frac{f_0}{1\ \text{Hz}}\right),\ \log_{10}\left(\frac{C_{\text{pad}}}{1\ \text{F}}\right),\ b\right]. \tag{12}$$

where $b = 0$ denotes L_{LC} and $b = 1$ denotes L_{CL}. Vendor selections, exhaustive-search results, and pass/fail labels are not used as model inputs.

At inference, the predicted residuals define the continuous vendor-search center:

$$L_{\text{pred}} = L_{\text{ideal}} e^{\widehat{\Delta}_L},\ C_{\text{pred}} = C_{\text{ideal}} e^{\widehat{\Delta}_C}. \tag{13}$$

Thus, the regressor localizes the vendor-search region but does not directly select the final component pair. Training and verification datasets are described in **Section III-A**.

C. Safe Vendor Library Construction

For each f_0, the raw vendor tables are filtered into a frequency-dependent safe library using the criteria in **Table I**.

TABLE I. Safe-Library Filtering Criteria Used for Vendor Component Selection

Filter criterion	Rejection condition	Purpose of test
Frequency coverage	Vendor data do not cover f_0	Ensures interpolation at the operating frequency
Numerical validity	Interpolated impedance is non-finite	Removes unusable vendor entries
Resistance validity	Negative resistance beyond numerical tolerance	Removes physically invalid component data
Resistance limit	$\Re\{Z\}>1000\ \Omega$	Removes highly lossy component regions
Impedance magnitude limit	$\lvert Z\rvert>2500\ \Omega$	Removes extreme impedance values in the 50 Ω matching benchmark
Self-resonance margin	$f_0>0.9f_{zc}$	Avoids operation near the first reactance zero-crossing

The first reactance zero-crossing frequency (f_{zc}) is estimated from the first sign change of the interpolated reactance curve. These filters avoid highly lossy, numerically unstable, or near-self-resonant regions; if no reactance zero crossing is observed, the zero-crossing margin filter is not applied. Negative real resistance is rejected only when $\Re\{Z\} < -10^{-6}\ \Omega$.

D. Local Vendor Search

Given $\left(L_{\mathrm{pred}}, C_{\mathrm{pred}}\right)$, safe vendor values are ranked by logarithmic distance from the corrected center:

$$d_L(L_i) = \left|\ln L_i - \ln L_{\mathrm{pred}}\right|, \quad d_C(C_j) = \left|\ln C_j - \ln C_{\mathrm{pred}}\right| \quad (14)$$

For a $k \times k$ search, the (k) nearest safe inductor values and (k) nearest safe capacitor values define

$$\mathcal{N}_{k\times k} = \{(L_i, C_j) : L_i \in \mathcal{L}_k,\ C_j \in \mathcal{C}_k\} \quad (15)$$

If fewer than (k) safe values are available along either axis, all available values are used. If the predicted center lies outside the safe-library range, the nearest boundary values are used rather than extrapolating.

Each pair in ($\mathcal{N}_{k\times k}$) is evaluated using the vendor-model circuit equations (2)–(6). The selected pair is

$$(L^*, C^*) = \arg \min_{(L_i, C_j)\in \mathcal{N}_{k\times k}} S_{11,\mathrm{dB}}(L_i, C_j) \quad (16)$$

The fixed deployable search uses (k=5), corresponding to at most 25 vendor-pair evaluations. A 7×7 search is used for fallback and sensitivity analysis.

E. Adaptive Fallback

The fixed 5×5 search may miss vendor-feasible pairs when the predicted center is distant from the feasible region. For feasible-frontier accounting, local no-pass cases are expanded to 7×7, followed by exhaustive safe-library verification when still unresolved. **Table IV** reports the staged cost over the exhaustive-feasible subset. In deployment, all local no-pass cases require the same expansion because feasibility is unknown beforehand.

III. Experimental Setup

A. Training and Verification Datasets

Verification used a held-out physics-guided synthetic dataset for low-resistance loads satisfying $R_L < Z_0$, with $Z_0 = 50\ \Omega$. The design range was $R_L = 6$–$45\ \Omega$, $f_0 = 0.8$–6.0 GHz, and $C_{\mathrm{pad}} = 1$–300 fF. Load reactance was generated as

$$\frac{X_L}{Z_0} \sim \mathcal{U}\left[-0.75\sqrt{r - r^2}, 0.75\sqrt{r - r^2}\right], \quad r = \frac{R_L}{Z_0}. \quad (17)$$

This bound retains the loads within the considered low-resistance synthesis region. The verification set contained 20,000 branch-specific cases generated from 10,000 unique source-load conditions, with 10,000 L_{LC} and 10,000 L_{CL} cases. The residual-learning dataset contained 14,880 branch-balanced samples, divided into 11,824 training, 1,515 validation, and 1,541 test samples using source-level splitting with no overlap. Dataset and model configurations are summarized in **Table III**.

B. Vendor Impedance Tables

Vendor evaluation uses Murata RF-inductor and multilayer ceramic-capacitor data obtained from Murata's design-support database [13], [14]. The frequency-dependent complex impedance data enable direct vendor-model recomputation during verification. The inductor table contains 6,416 impedance rows across 16 nominal inductance values, while the capacitor table contains 6,817 impedance rows across 17 nominal capacitance values. The nominal component ranges are 0.12–27 nH for inductors and 1.0–39 pF for capacitors, with frequency coverage of 0.05–20 GHz and 0.10–30 GHz, respectively.

For every verification case, component impedances are interpolated at f_0, filtered using the safe-library rules in **Section II-C**, and evaluated using the circuit model in **Section II-A**. Since the safe library depends on frequency coverage and self-resonance filtering, the number of valid vendor pairs varies across cases. Across the full verification set, the median exhaustive safe-library size is 240 candidate pairs.

C. Compared Methods

The primary deployable method is No-Q LMatchNet-V with a fixed 5×5vendor search. Analytical NN selects the nearest safe pair to the analytical seed, while analytical 5×5searches its local neighborhood. Random safe-25 uses one fixed-seed realization. Elite-guided safe-25 begins with seven random pairs, then samples unrepeated candidates around elite observations and returns the best vendor-modeled $S_{11,\mathrm{dB}}$after 25 evaluations. The adaptive No-Q result uses feasible-frontier accounting. A fixed $k \times k$search costs $O(k^2)$, whereas exhaustive search costs $O(N_{\mathrm{safe}})$.

TABLE II. Compared vendor-search policies and evaluation budgets

Method	Evaluation budget	Purpose of test
Analytical NN	1	Direct rounding of analytical seed
Analytical 5×5	25	Local search around analytical seed
Random safe-25	25	Uninformed budget-matched search
Elite-guided safe-25	25	Budget-matched elite-guided stochastic search
No-Q 5×5	25	Proposed deployable fixed search
No-Q adaptive	Variable; median 25 under frontier accounting	Feasible-frontier recovery with fallback
Exhaustive	N_{safe}	Vendor-feasible upper bound

D. Model and Feature Audits

Preliminary audits compared ridge, elastic-net, k-nearest-neighbor, random-forest, extra-trees, gradient-boosting, HGBR, and multilayer-perceptron regressors [6], [7], [9]. HGBR was retained based on validation performance, while the deployable inputs were restricted to R_L/Z_0, X_L/Z_0, f_0, C_{pad}, and branch. Synthetic Q_{ctrl}, vendor selections, exhaustive-search results, and pass/fail labels were excluded from the inputs. The final no-Q configuration is summarized in **Table III**.

E. Evaluation Protocol

All pass/fail decisions are recomputed directly from the interpolated vendor impedances using the reflection-coefficient and $S_{11,\mathrm{dB}}$ calculation in (6); stored pass/fail labels are not used as ground truth. A design is considered feasible when exhaustive safe-library evaluation finds at least one candidate satisfying $S_{11,\mathrm{dB}} \le -10$ dB. The primary

metric is feasible-frontier recovery, defined as the fraction of exhaustive-feasible cases recovered by each evaluated method; exhaustive-infeasible cases are excluded from the denominator.

TABLE III. MODEL, TRAINING AND SEARCH CONFIGURATION

Category	Configuration
Learner	Two HGBR residual regressors
Inputs/Targets	$\left[\frac{R_L}{Z_0}, \frac{X_L}{Z_0}, \log_{10}(f_0), \log_{10}(C_{\text{pad}}), b\right] \rightarrow (\Delta_L, \Delta_C)$
Primary training range	R_L = 6–45 Ω, f_0 = 0.8–6.0 GHz, C_{pad} = 1–300 fF, $\frac{X_L}{Z_0}$ from (17)
Training set	14,880 branch-balanced samples; 11,824/1,515/1,541 train/validation/test split
HGBR settings	450 iterations; learning rate 0.04; max leaf nodes = 31; $L_2 = 0.01$; seed 42
Vendor data	Murata RF tables: 6,416 L rows and 6,817 C rows; 16 L and 17 C nominal values
Safe library	Median 240 vendor pairs after feasibility filtering
Pass criterion	$S_{11,\text{dB}} \leq -10$ dB
Primary search	no-Q 5×5 local search, at most 25 evaluations
Benchmark	Exhaustive safe-library search

Evaluation cost is measured by the number of unique vendor pairs recomputed. A case terminating after the 5×5 stage requires at most 25 evaluations, while termination after the 7×7 stage requires at most 49 cumulative evaluations. If exhaustive fallback is required, the cumulative count is N_{safe}, because candidates evaluated during the local stages are reused rather than counted again. Relative to the median 240-pair safe library, the fixed 5×5 and 7×7 searches reduce the evaluated candidate count by 89.58% and 79.58%, respectively. Their median runtimes are 4.4 and 6.1 ms per case, compared with approximately 26 ms for exhaustive search.

IV. RESULTS AND DISCUSSION

A. Self-Contained Evaluator Sanity Check

The vendor evaluator was validated against saved 5×5 verification results for 500 randomly selected cases. Pass/fail decisions and safe-library counts showed 100.00% agreement, and recomputed S_{11} values matched the saved results to numerical precision. A 2,000-case sensitivity sweep over zero-crossing margins of 80%, 90%, and 95%, and $|Z|$ cutoffs of 1500, 2500, and 5000 Ω, was performed as an implementation sanity check for the safe-library filters.

B. Main No-Q Verification Result

Table IV reports standalone costs for the fixed 5×5 and 7×7searches, and cached cumulative evaluation counts for the adaptive $5 \times 5 \rightarrow 7 \times 7 \rightarrow$ exhaustive procedure over exhaustive-feasible cases.

Recovery was evaluated over the 5,514 exhaustive-feasible cases using two-sided 95% Wilson intervals. The fixed $5 \times 5$and 7×7searches recovered 4,878 cases (88.47%) and 5,066 cases (91.88%) using at most 25 and 49 evaluations, respectively. Of the 636 cases missed initially, 188 were recovered by the wider search and 448 required exhaustive fallback. Under feasible-frontier accounting, adaptive verification recovered all 5,514 feasible cases with a median of 25 unique evaluations.

TABLE IV. MAIN NO-Q VERIFICATION RESULTS

Method	Recovery	95% Wilson CI	Median evals	Time med./mean (ms/case)
No-Q fixed 5×5	4878/5514 (88.47%)	87.60–89.28%	25	4.41/5.02
No-Q fixed 7×7	5066/5514 (91.88%)	91.12–92.57%	49	6.11/7.01
No-Q Adaptive	5514/ 5514 (100.00%)	99.93–100.00%	25	-

C. Comparison with Baselines

Analytical nearest-neighbor recovered 71.60% of the feasible frontier using one evaluation. Under the same 25-evaluation budget, random safe-25, analytical-centered search, Elite-guided safe-25, and the proposed no-Q search achieved 83.42%, 86.33%, 89.37%, and 88.47% recovery, respectively. The proposed method therefore improved upon random and analytical localization, while remaining within 0.90 percentage points of the Elite-guided baseline. Its main distinction is deterministic physics-seeded localization followed by vendor-model verification and adaptive fallback.

Fig. 2(a) shows the empirical CDF (ECDF) of achieved S_{11} over exhaustive-feasible cases, where the ECDF gives the fraction of cases whose S_{11} is at or below each value, with the dashed line indicating the -10 dB pass threshold. **Fig. 2(b)** presents the survival curve of $\Delta S_{11} = S_{11,\text{method}} - S_{11,\text{exhaustive}}$, where smaller values indicate closer agreement with the exhaustive safe-library bound.

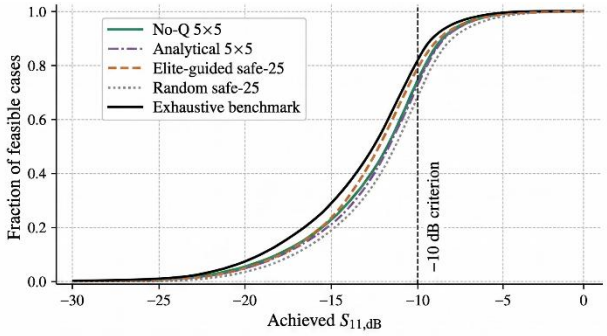


Fig. 2(a). CDF of achieved S_{11}—absolute matching performance.

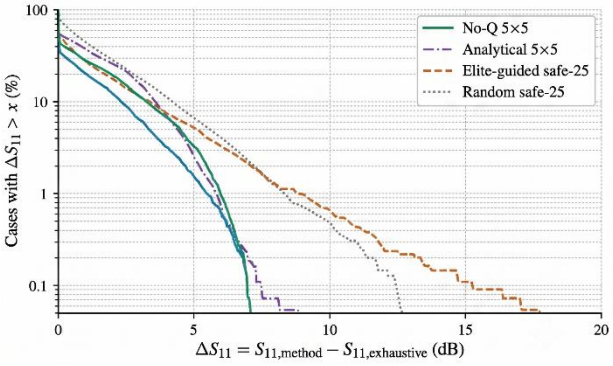


Fig. 2(b). Survival of excess S_{11} gap—deviation from exhaustive search.

Fig. 2. Vendor-table performance over the 5,514 exhaustive-feasible cases: (a) empirical CDF of achieved $S_{11,\text{dB}}$, with the dashed line indicating the -10 dB matching criterion; and (b) survival function of the excess gap $\Delta S_{11} = S_{11,\text{method}} - S_{11,\text{exhaustive}}$.

D. Adaptive Fallback Behavior

Of the 5,514 exhaustive-feasible cases, the initial 5×5search recovered 4,878 cases. The 7×7expansion recovered 188 additional cases, while the remaining 448 required exhaustive safe-library fallback. Thus, adaptive expansion recovered the complete feasible frontier while retaining the low-cost local search for most feasible cases. In deployment, every local no-pass case proceeds to wider or exhaustive verification because its exhaustive feasibility is not known in advance.

E. Ablation and Branch Diagnosis

Table V reports branch-wise recovery for the no-Q vendor-table pipeline. The L_{LC}branch was fully recovered by both the $5 \times 5$and 7×7searches. All fixed-neighborhood misses occurred in the L_{CL} branch, with 636 misses for 5×5 and 448 misses for 7×7 . Thus, the series-capacitor/shunt-inductor branch was more difficult for learned search localization and benefited more from fallback.

TABLE V. BRANCH-WISE NO-Q RECOVERY DIAGNOSIS

Branch	Feasible	5×5 recovery	7×7 recovery	Note
L_{LC}	2056	2056/2056, 100.00%	2056/2056, 100.00%	fully recovered
L_{CL}	3458	2,822 / 3,458 81.61%	3,010 / 3,458 87.05%	dominant miss source

F. Discussion

LMatchNet-V is best interpreted as a search-localization framework rather than a direct component predictor [10]. The fixed no-Q 5×5 search evaluates only 25 vendor pairs, compared with a median safe-library size of 240 pairs, while recovering 88.47% of the exhaustive-feasible frontier. Residual learning improves recovery by 2.14 percentage points over analytical-centered search without increasing the evaluation budget. Expanding the neighborhood to 7×7raises recovery to 91.88%, and adaptive fallback recovers the remaining feasible cases through verified vendor-table evaluation.

The learned residual corrects systematic displacement introduced by pad capacitance and the finite-Q target model, while the subsequent local vendor search handles catalog quantization and frequency-dependent component behavior. The synthetic Q_{ctrl} variable is used only during target generation and is excluded from deployment inputs. Final component selection is therefore determined by the interpolated vendor impedance model rather than by the regressor itself.

Table VI combines generated boundary stress sets with outcome-defined diagnostic subsets. The near-$50\ \Omega$ and operating-endpoint rows test extrapolation, whereas the feasibility-frontier and library-boundary rows diagnose difficult regions of the evaluated solution space.

As shown in **Table VI**, the fixed 5×5search maintains strong recovery for operating-endpoint (82.68%) and feasibility-frontier (98.25%) cases, but degrades for near-50 Ω loads (43.28%) and library-boundary cases (38.25%). Near Z_0, the required reactance approaches zero, yielding a sparse feasible region, while parasitic effects at catalog boundaries shift the optimal discrete solution away from the analytical seed. Nevertheless, the median evaluation count remains 25 across all regimes, demonstrating bounded search cost.

TABLE VI. BOUNDARY STRESS SETS AND DIAGNOSTIC SUBSETS

Load/Test Regime	Total / Exhaustive-Feasible	5×5 recovery (95% CI)	7×7 recovery (95% CI)
Existing Low-R_{L} Cases (Primary)	20,000 / 5,514	88.47% (87.60–89.28%)	91.88% (91.12–92.57%)
Near 50Ω Upper Boundary (R_{L} =45.1–49.9 Ω) (beyond training range)	400 / 134	43.28% (35.20–51.74%)	58.21% (49.74–66.22%)
Operating Endpoints (f_0, C_{pad} extremes)	400 / 179	82.68% (76.47–87.52%)	86.59% (80.83–90.82%)
Feasibility Frontier (≈ −10 dB)	171 / 171	98.25% (94.97–99.40%)	100.00% (97.80–100.00%)
Library Boundaries (min/max L or C)	400 / 400	38.25% (33.62–43.10%)	47.50% (42.65–52.39%)

A preliminary cross-library study examined whether the search-localization approach remained effective beyond the primary Murata library. Using a separate model configuration and the same 25-pair search budget, feasible-frontier recovery reached 99.32% for Murata LQW15/GJM, 93.53% for Murata LQW18/GJM, and 92.35% for the Coilcraft 0402DC [15] with KYOCERA AVX Accu-P 0402 [16]. The corresponding median safe-library sizes were 240, 600, and 4,770 pairs, respectively. These results provide supporting evidence that physics-seeded localization can remain effective across component families and vendors. Because this experiment used a separate model configuration, it is treated as a preliminary transfer study and is not directly compared with the locked no-Q results in **Table IV**.

The present validation is limited to circuit-level vendor-model simulation using Murata impedance tables and synthetically generated load states. Hardware measurement, component tolerances, PCB layout parasitics, and full-wave electromagnetic validation remain outside the scope.

TABLE VII. COMPARISON OF LMATCHNET-V WITH REPRESENTATIVE RF MATCHING AND OPTIMIZATION METHODS

Ref.	Method and component treatment	Selection mechanism / distinction
[2], [3], [11]	Learning-based matching using simulated or tunable values	Direct value prediction or control; no finite vendor-catalog verification
[10]	Adaptive L-network with tunable elements	Control-based adjustment; no learned catalog localization
[17]	GA-based WPT matching with discrete passives and switching	Iterative evolutionary component search
[18]	SMT matching with component and board parasitics	Discrete BOM selection through EM–circuit synthesis; no learned residual localization
[19]	Parasitic-aware data-driven tunable matching	Learned surrogate with numerical or inverse optimization; no vendor-catalog post-synthesis mapping
This work	**Physics-seeded residual learning with a finite vendor catalog**	**ML localizes the search; vendor-model recomputation selects the final pair**

Table VII positions LMatchNet-V as a post-synthesis vendor-selection workflow. Unlike direct value prediction or tunable-control methods, ML only localizes a finite,

frequency-dependent catalog, while vendor-model recomputation retains final selection authority.

G. Ablation Study

To quantify the contribution of each framework component, a stepwise ablation was performed on the 20,000-case set. The results are summarized in **Table VIII**.

TABLE VIII. WITHIN-FRAMEWORK ABLATION RESULTS

Component	Method	Recovery	Evaluations	Gain vs. Previous
Physics seeding	Analytical NN (1 eval)	71.60%	1	—
Local search	Analytical 5 × 5 (25 evals)	86.33%	25	+14.73 pp
Residual learning	No-Q 5 × 5 (25 evals)	88.47%	25	+2.14 pp
Wider search	No-Q 7×7 (49 evals)	91.88%	49	+3.41 pp

Within the proposed framework, physics seeding provides the foundation, while local search captures the discrete neighborhood structure and residual learning further refines the search location without increasing the evaluation budget. Expanding the neighborhood improves recovery at additional cost, confirming that the analytical seed, learned correction, and vendor-search stages provide complementary gains in vendor-component localization.

V. CONCLUSION

This paper presented LMatchNet-V, a physics-seeded residual-learning framework for vendor-aware single-frequency L-section impedance matching. Classical synthesis provides a continuous component seed, while the learned residual localizes a compact region of the discrete vendor library. Final component selection and pass/fail verification remain governed by frequency-dependent vendor-model recomputation rather than direct ML prediction.

The framework was evaluated on 20,000 branch-specific cases, of which 5,514 were feasible under exhaustive safe-library search. The fixed no-Q 5×5 search recovered 88.47% of the feasible frontier using 25 vendor-pair evaluations, while the 7×7search recovered 91.88% using 49 evaluations. Under feasible-frontier accounting, adaptive expansion and exhaustive fallback recovered all feasible cases when complete recovery was required. These results demonstrate a controllable trade-off between evaluation budget and vendor-feasible recovery.

The present study is limited to synthetic load states, single-frequency operation, and circuit-level vendor-model evaluation. Future work will include fabricated-board and VNA validation, component tolerances, PCB and electromagnetic parasitics, broader vendor-library testing, CAD integration, and extensions to multi-section and broadband matching.

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